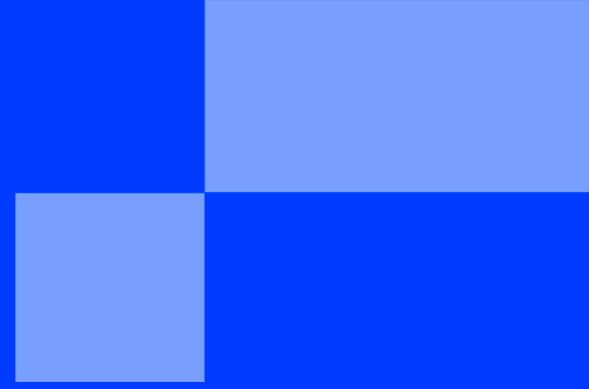
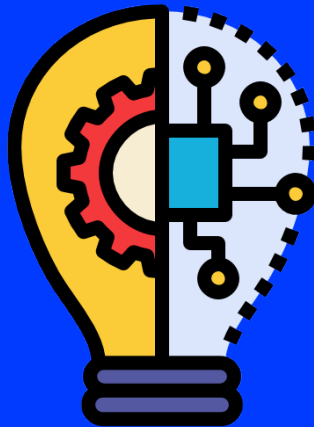




SOLUTION TRACK

< VIDEO GAME – TECH 4 />



SOLUTION TRACK

Introduction

Welcome to the Solution Track of the Epitech Innovative Project (EIP) – where technology meets real-world impact. This track empowers you to develop innovative video games that address genuine problems or deliver strong experiential value. Your mission is to create a game that doesn't just work technically, but resonates with players, solves meaningful play or design challenges, and builds lasting engagement.

The Solution Track for video games focuses on the human side of gameplay. While your code and engine work form the foundation, success here means understanding your players deeply, crafting experiences they love, and building a community that believes in your vision. You'll learn to think like a game designer, a community builder, and a player advocate – skills that transform good developers into exceptional game creators.

This journey will challenge you to master multiple dimensions of game development. You'll conduct player research to validate your assumptions, design and iterate on game experience based on real feedback, build and nurture an engaged player community, and optionally explore complementary objectives such as art direction, innovative mechanics, content ecosystems, distribution and monetization, or technical performance and accessibility. Each aspect is crucial for transforming a technical game project into a solution that truly matters to players.

This guide provides a comprehensive framework for your Video Game – Tech 4 Solution Track journey, detailing both **mandatory objectives** that form the core of your assessment and **complementary goals** you'll select with your mentor to match your project's unique needs. Clear deliverables and evaluation criteria ensure you understand what excellence looks like at every step.

Remember that alongside these solution-focused objectives, you'll continue advancing your technical development. The workload will ebb and flow throughout the year, with some periods demanding more intensive effort. Strong organizational skills and open communication with your mentor and the EIP academic coordinator will be your keys to success. They are here to challenge your assumptions, refine your strategies, and help you navigate the complex but rewarding path of building games that matter.

Mandatory Objectives to Achieve

1. Develop and Retain a Player Community

Identification and Targeting

- ✓ **Define one or more player personas:** gaming profiles (casual/hardcore), preferred genres, platforms, playtime habits, frustrations, and expectations.
- ✓ **Clarify the game value proposition:** why would players be interested in playing this game rather than another? What is unique or compelling?
- ✓ **Analyze competitive positioning:** benchmark 3–5 similar games, identify overlaps, gaps, and opportunities for differentiation.

Community Channel Setup and Management

- ✓ **Create and actively manage at least 2 appropriate channels** such as:
 - a Gaming Discord server, and
 - a store or community platform presence (Steam, itch.io, Reddit, Twitter/X, etc.).
- ✓ **Build a cohesive brand identity** for the game:
 - logo and key art,
 - communication tone of voice,
 - visual style and color palette,
 - consistent messaging across channels.
- ✓ **Define a game-focused content plan**, including for example:
 - weekly or bi-weekly devlogs,
 - gameplay GIFs and compelling screenshots,
 - progress or feature highlight videos (before/after, new features),
 - behind-the-scenes content (concept art, discarded prototypes),
 - patch notes and a public roadmap,
 - teasers and content reveals.

Playtester Acquisition and Activation

- ✓ **Implement a targeted playtester recruitment strategy** using:
 - gaming forums and thematic Discord servers,
 - Reddit communities,
 - schools or local communities,
 - existing network of players.
- ✓ **Reach a minimum of 30 identified, onboarded, and active playtesters.**
- ✓ **Create a structured playtest journey** including:
 - dedicated playtest builds,
 - feedback forms or surveys,
 - bug tracking process and tools.
- ✓ **Define clear gaming incentives:**
 - early access builds,
 - in-game credits or recognition,
 - exclusive cosmetic items or skins,
 - VIP roles or channels on Discord.

Tracking, Measurement, and Retention

- ✓ **Implement an engagement dashboard**, tracking at least:
 - average playtime,
 - retention rate,
 - completion rate or progression metrics,
 - reported bugs and issues.
- ✓ **Integrate player feedback** into sprint reviews and maintain a public changelog.
- ✓ **Organize community events** such as:
 - game jams or internal challenges,
 - speedrun contests,
 - feature votes or polls,
 - developer play sessions with the community.
- ✓ **Set up a reward system** for active contributors:
 - Easter eggs referencing them,
 - NPCs or items named after them,
 - exclusive in-game content.

Expected Deliverables (Non-exhaustive)

- ✓ **Detailed player persona sheets** (PDF, Miro, or similar).
- ✓ **Links to active Discord and store/community pages** (Steam/itch.io/etc.) with visible followers or members.
- ✓ **Editorial communication calendar** (Notion, Trello, spreadsheet).
- ✓ **Archive of devlogs and community posts.**
- ✓ **Playtest analytics** (heatmaps, funnels, retention metrics).
- ✓ **List of playtesters** with profiles and contributions.
- ✓ **Public bug tracker** (GitHub Issues, Trello, etc.).
- ✓ **Documented changelog** including community-attributed changes.
- ✓ **Professional press kit** (for the game and community outreach).
- ✓ **Compilation of player testimonials** (video or written).

What Is Not Sufficient

- ✓ Opening a Discord without regular activity, structure, or moderation.
- ✓ Posting a few GIFs without strategy or interaction.
- ✓ Showing “vanity metrics” (followers) without real engagement proof.
- ✓ Only testing with family, close friends, or internal team members.
- ✓ Ignoring negative or constructive feedback.
- ✓ Promising unrealistic features just to attract attention.

2. Optimize Game Experience (Game UX)

Analysis and Definition of Target Experience

- ✓ **Identify clear game design pillars:**
 - core gameplay loop,
 - main mechanics,
 - targeted player emotions and fantasy.
- ✓ **Create a structured Game Design Document (GDD)** describing:
 - vision and pillars,
 - systems and mechanics,
 - progression and pacing,
 - in-game economy if relevant.
- ✓ **Define key player moments:**
 - first-time experience (FTUE),
 - skill gates or mastery thresholds,
 - moments of climax,
 - endgame or long-term goals.
- ✓ **Benchmark 3–5 reference games in the genre** and extract UX best practices.

Playtest and Iteration Methodology

- ✓ **Organize at least 5 formalized playtest sessions** with:
 - defined protocols,
 - clear goals per session.
- ✓ **Mix methods** such as:
 - live observation,
 - “thinking aloud” protocols,
 - post-session questionnaires,
 - analytics-based observation (telemetry).
- ✓ **Ensure diversity of test profiles:**
 - genre novices,
 - experienced players,
 - non-gamers,
 - main target audience.
- ✓ **Document at least 2 major iteration cycles** with before/after comparisons and rationale.

Onboarding and Learning Curve

- ✓ **Design an integrated or contextual tutorial**, avoiding walls of text.
- ✓ **Ensure time to first significant action is under 30 seconds.**
- ✓ **Shape a progressive difficulty curve** with clearly identified skill gates.
- ✓ **Provide immediate, readable feedback** on player actions.
- ✓ **Include accessibility options** when relevant:
 - control remapping,
 - difficulty settings,
 - assists or aids for specific challenges.

Polish and Game Feel

- ✓ **Implement game “juice”** such as:
 - screenshake,
 - hit pause,
 - particle effects,
 - squash & stretch where appropriate.
- ✓ **Ensure coherent multi-sensory feedback:**
 - visuals,
 - audio,
 - haptic feedback where applicable.
- ✓ **Guarantee responsive controls:**
 - target input lag below 100ms on median hardware,
 - fluid and readable animations.
- ✓ **Maintain UI/UX consistency:**
 - visual hierarchy,
 - affordance and clarity,
 - consistent patterns and symbols.
- ✓ **Track game feel metrics** when possible:
 - reaction times,
 - required precision,
 - timing windows.

Balancing and Progression

- ✓ **Document the game economy** with formulas, curves, and assumptions.
- ✓ **Design multi-layered progression systems:**
 - player skill progression,
 - character or account progression (stats, skills, unlocks).
- ✓ **Offer meaningful choices:**
 - varied builds or playstyles,
 - multiple strategies or paths,
 - replayability hooks.
- ✓ **Adjust balance based on playtest data:**
 - win/loss rates,
 - completion times,
 - friction or drop-off points.

Expected Deliverables (Non-exhaustive)

- ✓ **Complete and versioned Game Design Document.**
- ✓ **Playtest reports with notes and/or annotated videos.**
- ✓ **Gameplay flowcharts and UI wireframes.**
- ✓ **Balancing tables** with key metrics (damage curves, difficulty ramps, etc.).
- ✓ **Vertical slice or playable demo** showcasing core experience.
- ✓ **Comparative video or documentation of iterations.**
- ✓ **Behavioral heatmaps and analytics dashboards.**
- ✓ **Accessibility documentation.**
- ✓ **UI/UX style guide** for the game.
- ✓ **Player progression matrix.**

What Is Not Sufficient

- ✓ A purely theoretical GDD never updated alongside production.
- ✓ Only doing internal tests without any protocol or external players.
- ✓ Adding features without playtest validation.
- ✓ Neglecting game feel in favor of adding more content.
- ✓ A 30-minute tutorial before “real” gameplay begins.
- ✓ Ignoring accessibility under the guise of being “hardcore.”

Complementary Objectives

(2 to be chosen by the group, in consultation with your mentor)

A. Create a Cohesive Artistic Identity

Art Direction Design

- ✓ **Create a complete art bible** including:
 - concepts and roughs,
 - color palettes,
 - stylistic references,
 - technical constraints (resolution, polycount, texture budgets, etc.).
- ✓ **Define a unique visual style:**
 - realistic, stylized, pixel art, low poly, etc.,
 - clear rationale for choices vs. gameplay and audience.
- ✓ **Produce thematic moodboards** by:
 - environment,
 - characters,
 - UI and HUD.
- ✓ **Analyze at least 5 visual references** and identify differentiating elements for your game.

Asset Production and Pipeline

- ✓ **Produce at least 70% original assets** used in the game (models, textures, animations, VFX, UI elements).
- ✓ **Document the asset production pipeline:**
 - workflow steps,
 - naming conventions,
 - formats and export settings,
 - optimization rules.
- ✓ **Check and enforce visual consistency** across all assets.
- ✓ **Manage variations:**
 - skins,
 - environment variants,
 - props and modular pieces.

Audio and Sound Direction

- ✓ **Design a coherent sound identity:**
 - music,
 - ambiences,
 - SFX,
 - voice (if applicable).
- ✓ **Layer sound design** (ambient, action, feedback, UI).
- ✓ **Ensure precise audio-visual synchronization** to reinforce game feel.
- ✓ **Adapt mixing and mastering** to the main medium (headphones, speakers, mobile).

Technical Optimization and Adaptation

- ✓ **Use LODs and asset optimization** adapted to each platform.
- ✓ **Design shaders and post-processing** in service of the intended style and readability.
- ✓ **Provide graphic scalability:**
 - low/medium/high/ultra presets,
 - resolution and performance options.
- ✓ **Define a performance budget by asset type** (triangles, textures, draw calls, etc.).

Expected Deliverables (Non-exhaustive)

- ✓ **Complete art bible** (PDF, website, or online board).
- ✓ **Moodboards and concept sheets.**
- ✓ **Timelapse or breakdown of key asset creation.**
- ✓ **Visual effects showcase reel.**
- ✓ **Original soundtrack** (playlist, SoundCloud/Spotify, etc.).
- ✓ **Technical art documentation.**
- ✓ **With/without post-processing comparisons.**
- ✓ **Style guide for future assets and collaborators.**
- ✓ **“Behind the scenes” content** on art production.

What Is Not Sufficient

- ✓ Mixing assets from incompatible styles without adaptation / Using only store assets without changes.
- ✓ Neglecting audio in favor of visuals.
- ✓ Copying an existing style with no personal contribution.
- ✓ Over-using free VFX without integration work.
- ✓ Ignoring optimization with a “we’ll see later” mindset.

B. Implement Innovative Mechanics

Exploration and Ideation

- ✓ **Identify a design problem to solve** or a new emotion to create through gameplay.
- ✓ **Use structured brainstorming methods:**
 - mind maps,
 - internal game jams,
 - paper or low-fidelity prototypes.
- ✓ **Analyze unexplored spaces within your game genre.**
- ✓ **Document the “why”** behind each proposed innovative mechanic.

Rapid and Iterative Prototyping

- ✓ **Develop 3–5 different mechanic prototypes** exploring distinct directions.
- ✓ **Adopt a fail-fast approach:**
 - run quick tests,
 - justify abandonment of non-viable ideas,
 - capture learnings.
- ✓ **Document what each prototype taught you.**
- ✓ **Reuse “failed” ideas creatively** in other contexts if possible.

Integration and Synergy

- ✓ **Ensure the innovative mechanic is integrated** into the core loop, not a disconnected mini-game.
- ✓ **Create synergies with other systems** (progression, economy, narrative, multiplayer).
- ✓ **If hybridizing genres**, ensure transitions are fluid and narratively justified.
- ✓ **Encourage emergent but welcome strategies** that arise from system interactions.

Refinement and Polish

- ✓ **Move from concept to polished execution** for at least one key mechanic.
- ✓ **Fine-tune** timing, curves, feedback, and edge cases.
- ✓ **Aim for simplicity in learning, depth in mastery.**
- ✓ **Provide technical documentation** for implementation (for future contributors).

Expected Deliverables (Non-exhaustive)

- ✓ **Design document of innovative mechanics.**
- ✓ **Videos of chronological prototypes.**
- ✓ **Post-mortem of abandoned mechanics** (what was tried, why it was dropped).
- ✓ **Comparison with existing mechanics** from reference games.
- ✓ **Flowchart of interconnected systems.**
- ✓ **Commented source code excerpts for core mechanics.**
- ✓ **Playtester testimonials** about perceived innovation.
- ✓ **“GDC-style” presentation** on the main mechanic.

What Is Not Sufficient

- ✓ Innovating just to innovate, without real added value.
- ✓ Unnecessarily complicating existing mechanics.
- ✓ Copying a known mechanic and only changing minor parameters.
- ✓ Not testing mechanics with real players.
- ✓ Feature creep: constantly adding mechanics without consolidating.
- ✓ No documentation on design choices and trade-offs.

C. Develop a Content Ecosystem

Modular and Extensible Architecture

- ✓ **Design an architecture that supports content additions:**
 - DLC,
 - mods,
 - updates or seasonal events.
- ✓ **Separate core systems from content** using data-driven design.
- ✓ **Manage versioning and compatibility:**
 - save files,
 - mods,
 - multiplayer compatibility.
- ✓ **Provide technical documentation** focused on extensibility.

Community Creation Tools

- ✓ **Develop an integrated or standalone level editor** or content creation tool.
- ✓ **Offer mod support:**
 - SDK or API,
 - documentation,
 - sample projects.
- ✓ **Integrate content platforms** (Steam Workshop, mod.io) when relevant.
- ✓ **Provide templates and assets** to help creators get started.

Additional Content Pipeline

- ✓ **Design a post-launch content roadmap** over 6–12 months with clear milestones.
- ✓ **Plan seasonal content:**
 - events,
 - challenges,
 - rewards.
- ✓ **Implement automated systems** if relevant:
 - daily/weekly challenges,
 - rotating content.
- ✓ **Track content consumption metrics** (uptake, completion, retention).

Community Creative Engagement

- ✓ **Organize creation contests:**
 - maps,
 - skins,
 - mods or challenges.
- ✓ **Highlight community creations** with:
 - featuring in-game or on social channels,
 - rewards or recognition.
- ✓ **Co-create content with the community:**
 - votes,
 - suggestion boards,
 - beta content with feedback loops.
- ✓ **Attribute and credit contributors** transparently.

Expected Deliverables (Non-exhaustive)

- ✓ **Functional and documented level editor or creation tool.**
- ✓ **SDK/API with technical documentation.**
- ✓ **Examples of mods or custom content created by the team.**
- ✓ **Visual post-launch content roadmap.**
- ✓ **Configured Workshop/mod.io (or equivalent) page.**
- ✓ **Usage metrics for created content.**
- ✓ **Tutorial videos or guides** for creators.
- ✓ **License and guidelines** for modding and content creation.
- ✓ **Showcase of best community creations.**

What Is Not Sufficient

- ✓ Promising modding without providing actual tools.
- ✓ Providing an unstable or undocumented editor.
- ✓ Sharing an unrealistic or never-respected roadmap.
- ✓ Ignoring community-created content.
- ✓ Designing a closed architecture that prevents extension.
- ✓ Announcing DLC that is never delivered.

D. Distribution and Monetization Strategy

Market Analysis and Positioning

- ✓ **Study the target market:**
 - size,
 - trends,
 - major players,
 - opportunities.
- ✓ **Benchmark competitors:**
 - price,
 - key features,
 - reviews,
 - commercial performance.
- ✓ **Define unique positioning:**
 - genre,
 - target audience,
 - platforms,
 - price range.
- ✓ **Identify optimal release windows:**
 - avoid major AAA releases,
 - consider seasonality and events.

Multi-platform Presence

- ✓ **Create optimized store pages** (Steam, itch.io, Epic Store, GOG, etc., as relevant).
- ✓ **Run an active wishlist campaign**, aiming for at least 1000 wishlists before release when realistic.
- ✓ **Prepare builds for each target platform:**
 - PC, Mac, Linux, Web, etc.
- ✓ **Provide a demo version** to support conversion if appropriate.

Marketing and Communication

- ✓ **Produce professional trailers**, for example:
 - 30-second teaser,
 - 90-second gameplay trailer,
 - launch trailer.
- ✓ **Build a complete press kit**:
 - HD screenshots,
 - key art,
 - logos,
 - fact sheet,
 - GIFs.
- ✓ **Reach out to press and influencers**:
 - prepare a list of at least 20 targeted contacts,
 - send adapted pitches.
- ✓ **Ensure presence at events**:
 - game jams,
 - festivals,
 - student showcases,
 - online events.

Adapted Economic Model

- ✓ **Justify your business model choice**:
 - premium,
 - F2P,
 - early access,
 - freemium.
- ✓ **Align pricing strategy** with content, competition, and perceived value.
- ✓ **If applicable, design ethical monetization**:
 - no pay-to-win,
 - transparent loot boxes or randomized systems.
- ✓ **Produce realistic financial projections**:
 - costs,
 - revenues,
 - break-even scenarios.

Analytics and Optimization

- ✓ **Set up e-commerce analytics:**
 - wishlists,
 - conversions,
 - refunds.
- ✓ **Run A/B tests on store pages:**
 - screenshots,
 - descriptions,
 - tags and capsules.
- ✓ **Track business KPIs** where relevant:
 - CAC, LTV, ARPU,
 - retention and churn.
- ✓ **Plan a post-launch strategy:**
 - sales and bundles,
 - promotions,
 - live-ops events.

Expected Deliverables (Non-exhaustive)

- ✓ **Business plan with 12-month projections.**
- ✓ **Published and optimized store pages.**
- ✓ **Professional press kit** (presskit() or equivalent format).
- ✓ **Trailer(s) and marketing assets.**
- ✓ **List of press and influencer contacts.**
- ✓ **Analytics dashboard** (wishlists, traffic sources, conversions).
- ✓ **Marketing calendar** with milestones.
- ✓ **Compiled testimonials and reviews.**
- ✓ **Post-launch ROI analysis** (if applicable).

What Is Not Sufficient

- ✓ Releasing the game without any marketing efforts.
- ✓ Pricing disconnected from perceived value or competition.
- ✓ A sloppy Steam page with blurry or irrelevant screenshots.
- ✓ Forum or social media spam with no strategy.
- ✓ Aggressive monetization that alienates players.
- ✓ Ignoring the commercial dimension entirely.

E. Technical Performance and Accessibility

Performance Optimization

- ✓ **Target stable 60 FPS** on a clearly defined median configuration.
- ✓ **Run systematic profiling:**
 - CPU,
 - GPU,
 - memory,
 - I/O.
- ✓ **Provide graphic scalability:**
 - low to ultra presets,
 - multiple resolutions,
 - optional effects toggles.
- ✓ **Keep loading times below 30 seconds** on a standard HDD when possible.

Robust Technical Architecture

- ✓ **Implement a reliable save system:**
 - autosave,
 - cloud saves (if relevant),
 - versioning and migration strategy.
- ✓ **Handle errors gracefully:**
 - avoid crashes,
 - provide recovery paths,
 - integrate error reporting.
- ✓ **If multiplayer is present:**
 - optimize netcode,
 - handle lag compensation,
 - manage reconnections.
- ✓ **Implement basic anti-cheat measures** for competitive elements.

Structured Quality Assurance

- ✓ **Use systematic bug tracking:**
 - severity,
 - reproduction steps,
 - resolution status.
- ✓ **Introduce automated testing** where relevant:
 - unit tests,
 - integration tests,
 - smoke tests.
- ✓ **Test on multiple configurations:**
 - different hardware profiles,
 - stress and endurance tests.
- ✓ **Perform regression testing** before each release or major update.

Inclusive Accessibility

- ✓ **Offer visual options:**
 - colorblind modes,
 - UI scaling,
 - contrast adjustments.
- ✓ **Provide audio options:**
 - subtitles,
 - separate volume sliders,
 - visual cues for key sounds.
- ✓ **Support flexible controls:**
 - complete remapping,
 - one-handed modes,
 - hold/toggle options.
- ✓ **Design gameplay options:**
 - difficulty sliders,
 - assists,
 - skip options for specific sequences.
- ✓ **Align with recognized accessibility guidelines** (e.g., Game Accessibility Guidelines).

Technical Documentation

- ✓ **Write a complete README:**
 - requirements,
 - installation,
 - troubleshooting.
- ✓ **Maintain a detailed changelog** by version with fixes and improvements.
- ✓ **Provide a technical wiki** for modders and contributors if relevant.
- ✓ **Produce a technical post-mortem** summarizing learnings and key decisions.

Expected Deliverables (Non-exhaustive)

- ✓ **Benchmark report** on at least 5 configurations.
- ✓ **Profiler data** with documented optimizations.
- ✓ **Bug database** with status and resolution tracking.
- ✓ **Test coverage report** (for automated tests).
- ✓ **Accessibility audit checklist.**
- ✓ **Performance comparison** before/after optimization.
- ✓ **Detailed system requirements.**
- ✓ **Troubleshooting guide.**
- ✓ **Automated test suite** (where applicable).
- ✓ **Complete technical documentation** for future maintainers.

What Is Not Sufficient

- ✓ “Works on my machine” with no broader testing.
- ✓ Ignoring optimization until the last minute.
- ✓ Adding accessibility options without UX reflection or testing.
- ✓ No structured bug tracking.
- ✓ Minimal or non-existent technical documentation.
- ✓ Technical features that do not improve player experience.

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